

```
1 def merge_sort(arr):
2     if len(arr) <= 1:
3         return arr
4
5     mid = len(arr) // 2
6
7     left = merge_sort(arr[:mid])
8     right = merge_sort(arr[mid:])
9
10    return merge(left, right)
11
12
13 def merge(left, right):
14     result = []
15     i = j = 0
16
17     while i < len(left) and j < len(right):
18         if left[i] < right[j]:
19             result.append(left[i])
20             i += 1
21         else:
22             result.append(right[j])
23             j += 1
24
25     result.extend(left[i:])
26     result.extend(right[j:])
27
28     return result
29
30
31 # Driver code
32 arr = [38, 27, 43, 3, 9, 82, 10]
33
34 print("Original Array:", arr)
35 print("Sorted Array:", merge_sort(arr))
```

Original Array: [38, 27, 43, 3, 9, 82, 10]
Sorted Array: [3, 9, 10, 27, 38, 43, 82]

```
1 def binary_search(arr, key):
2     low = 0
3     high = len(arr) - 1
4     count = 0
5
6     while low <= high:
7         count += 1
8         mid = (low + high) // 2
9
10        if arr[mid] == key:
11            return mid, count
12        elif arr[mid] < key:
13            low = mid + 1
14        else:
```

```
15         high = mid - 1
16
17     return -1, count
18
19
20 # Driver code
21 arr = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
22
23 key = 70
24
25 position, iterations = binary_search(arr, key)
26
27 if position != -1:
28     print("Element found at index", position)
29     print("Number of iterations =", iterations)
30 else:
31     print("Element not found")
```

Element found at index 6
Number of iterations = 4